Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. We trained each model for 20 epochs with a batch size of 32.

2 Results

Figure 1 shows the test accuracy by dataset.

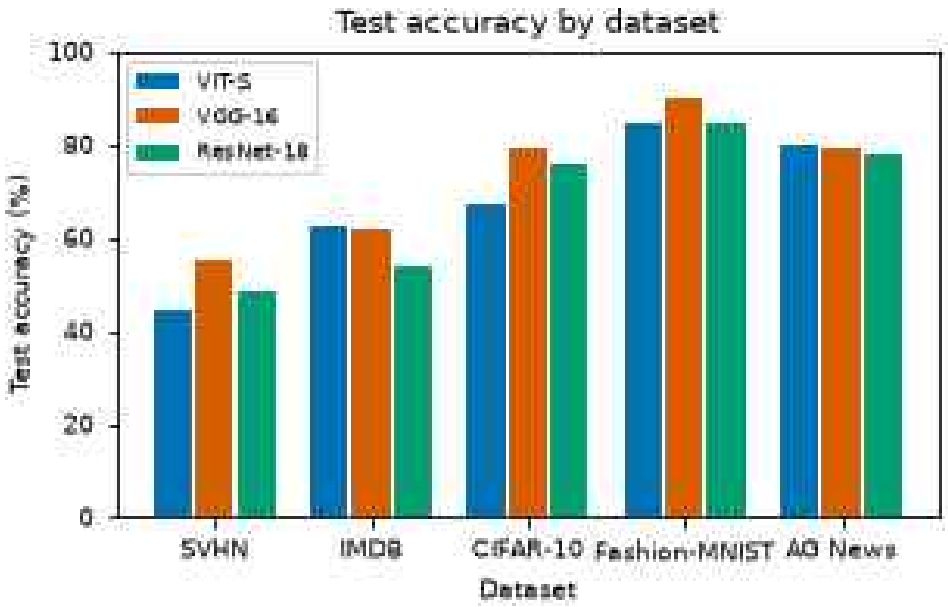


Figure 1: Test accuracy by dataset for the models in this experiment.

As shown in Figure 1, VGG-16 outperforms ResNet-18 by 55.1 percentage points on Fashion-MNIST. VGG-16 achieves 53.8% test accuracy on IMDB.

Figure 2 shows the validation accuracy by dataset.

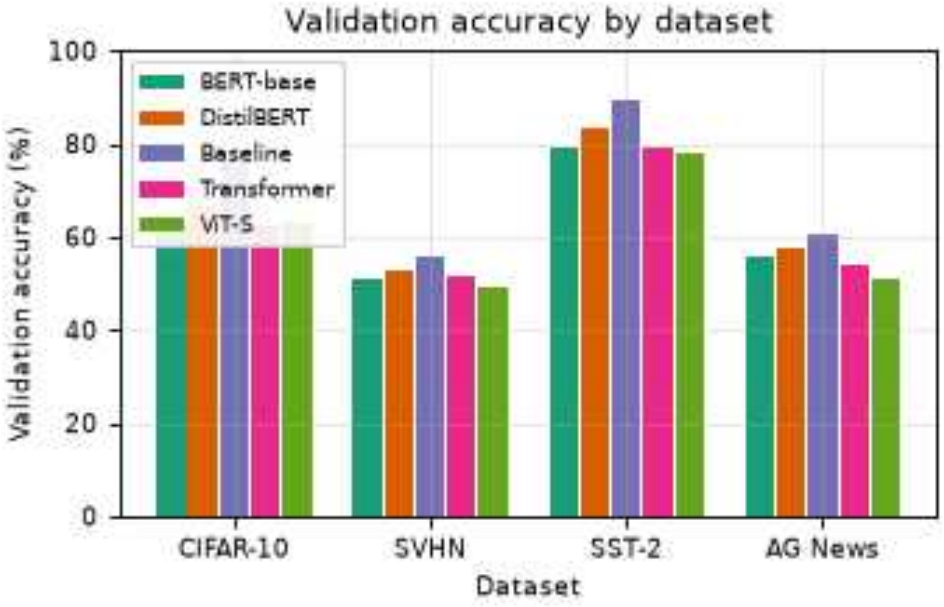


Figure 2: Validation accuracy by dataset for the models in this experiment.

On AG News, Baseline is 3.1 points better than DistilBERT (Figure 2). Transformer reaches over 75% validation accuracy on SST-2 (Figure 2).

Figure 3 shows the training loss.

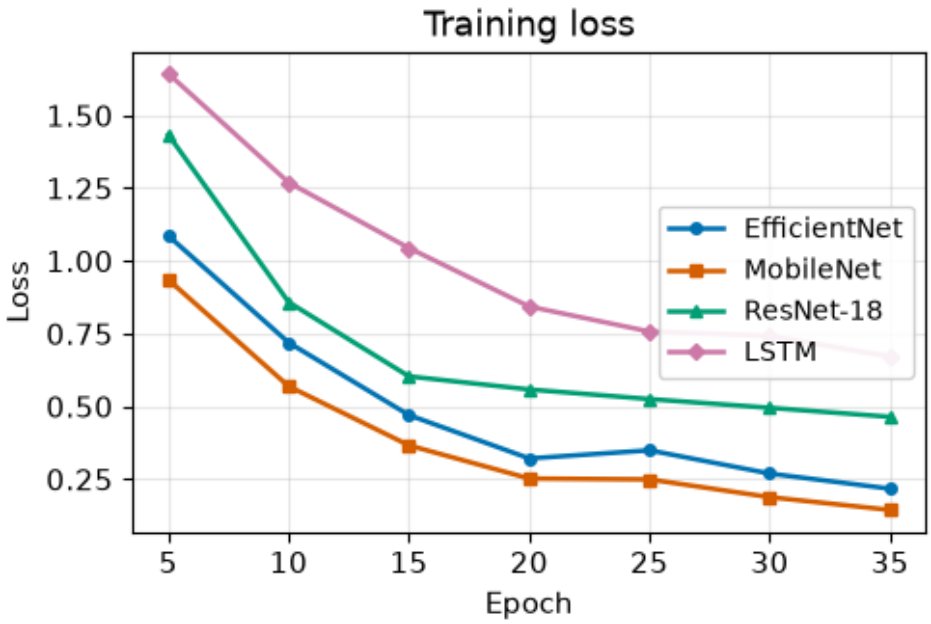


Figure 3: Training loss for the models in this experiment.

At epoch 25, the loss of MobileNet is 0.249 (Figure 3). As shown in Figure 3, at epoch 25, the loss of ResNet-18 is 0.349. As shown in Figure 3, EfficientNet has a loss of 0.104 at epoch 30.

3 Discussion

XGBoost reaches 52.5% accuracy on SST-2. Overall, the results are in line with what we expected from the literature.